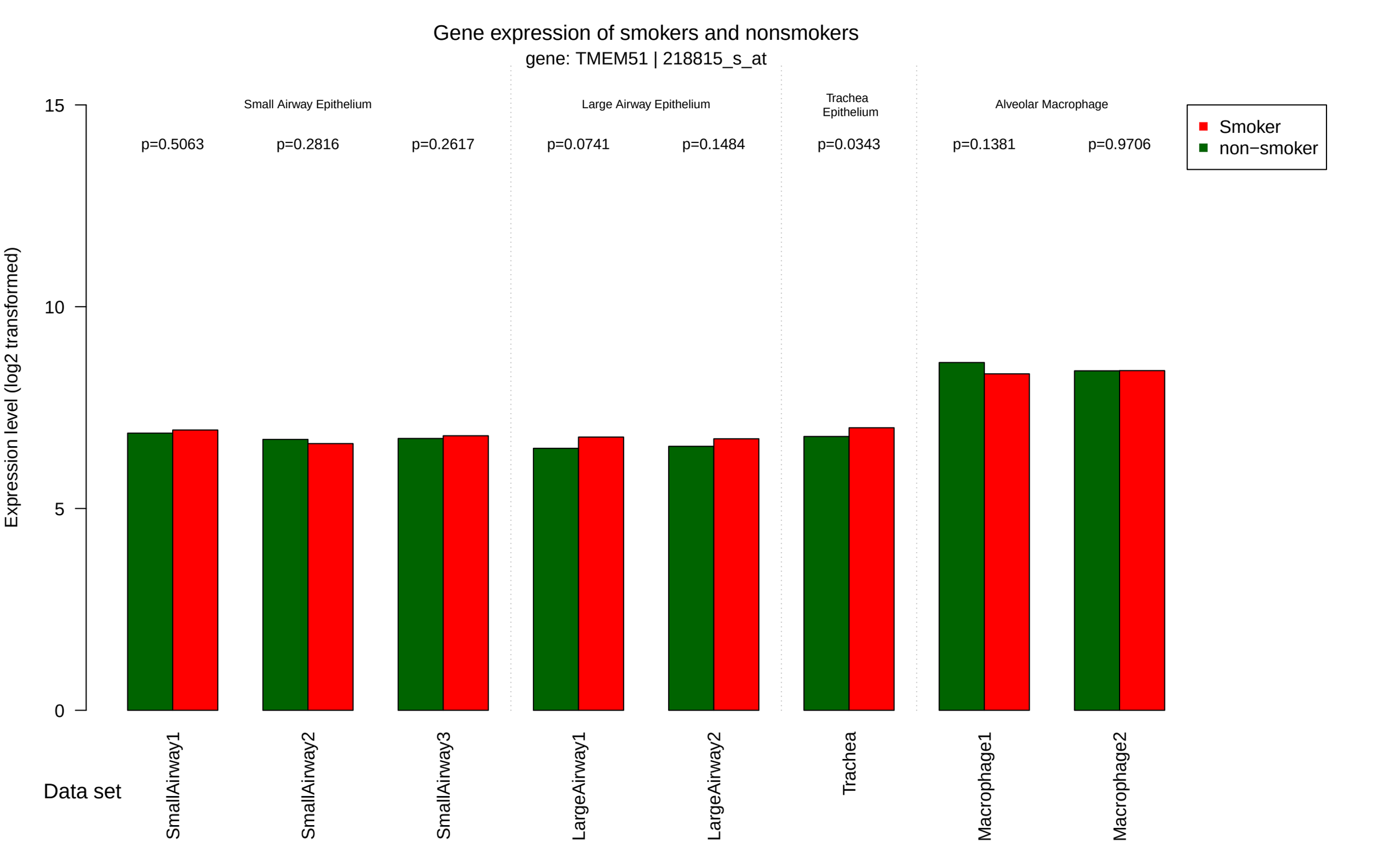
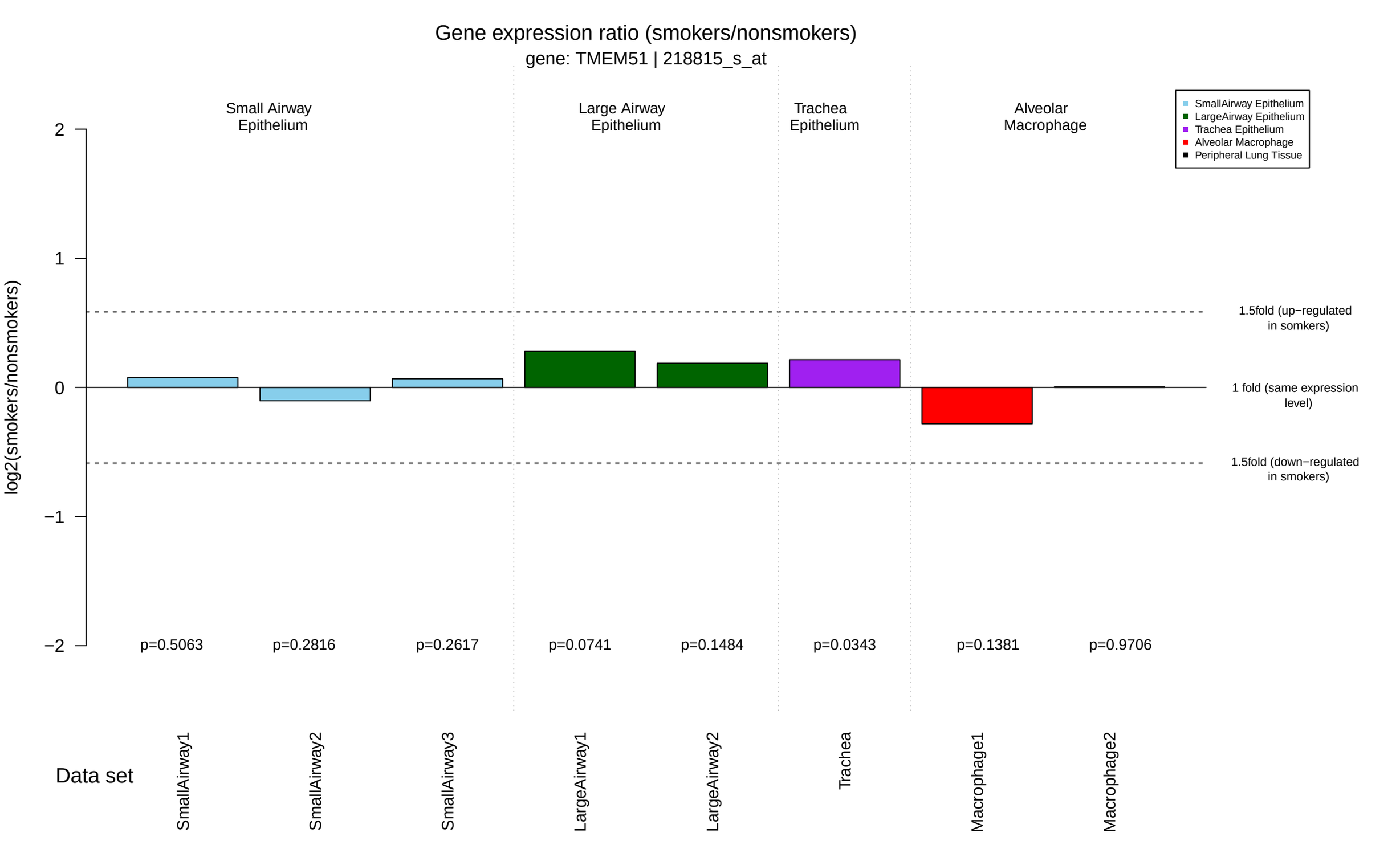
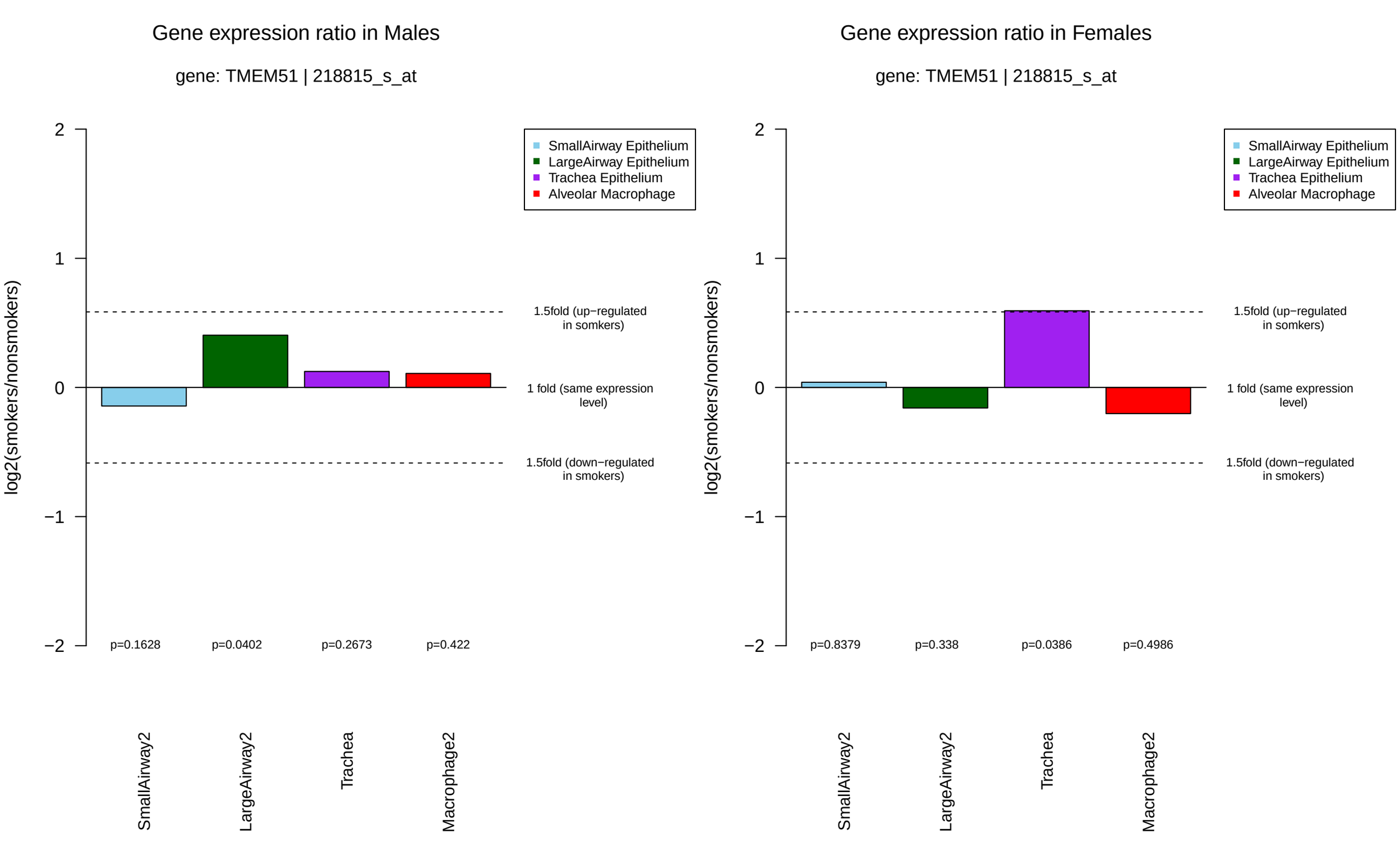
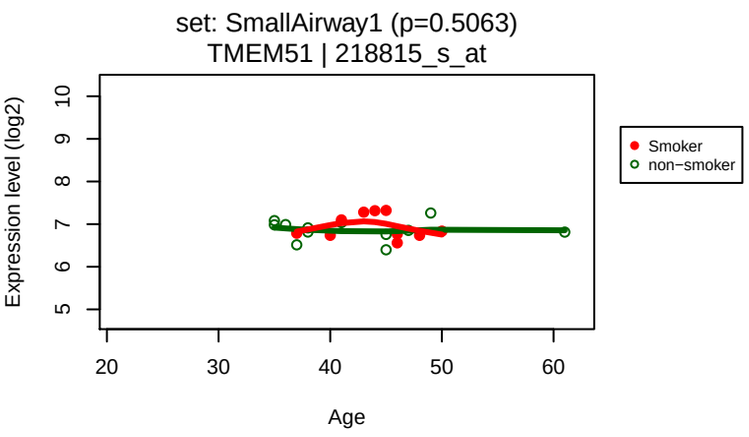
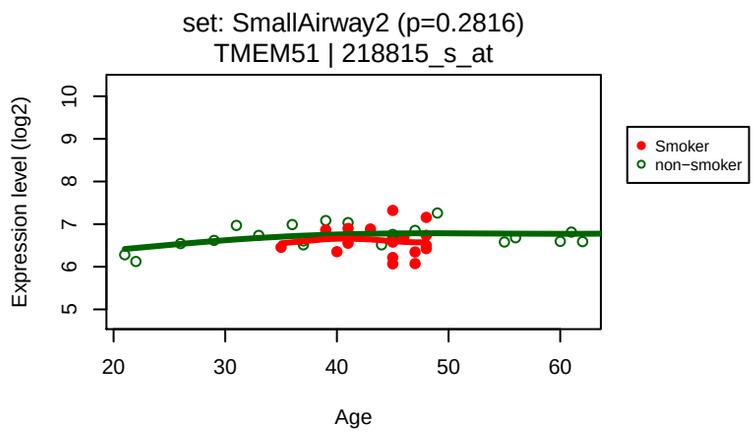
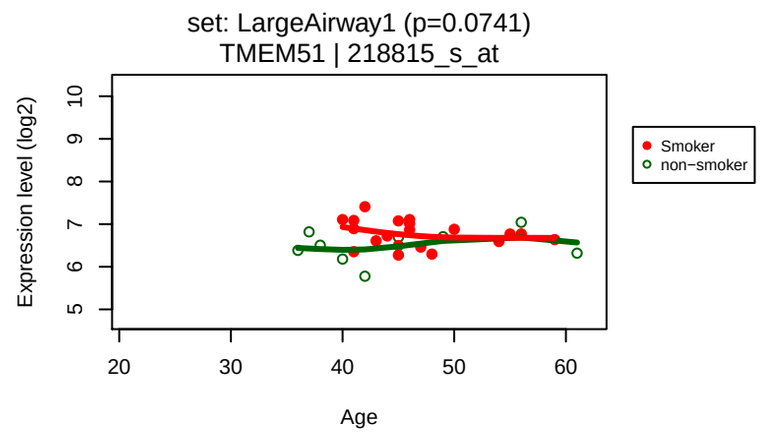

Gene expression of smokers and nonsmokers
gene: TMEM51 | 218815_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5063
p=0.2816
p=0.2617
p=0.0741
p=0.1484
p=0.0343
p=0.1381
p=0.9706
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

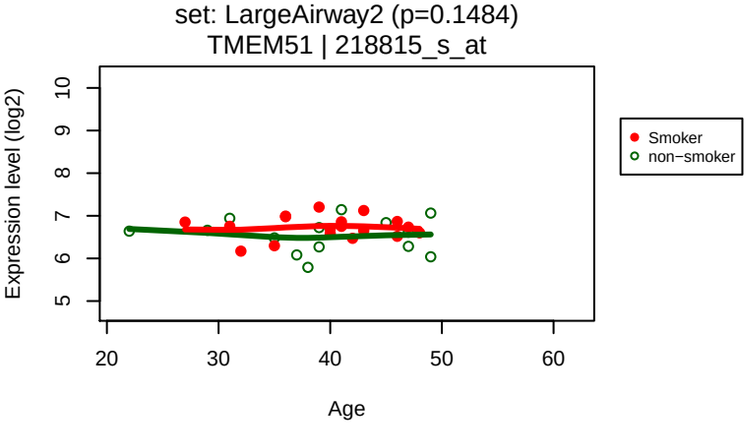

Gene expression ratio (smokers/nonsmokers)
gene: TMEM51 | 218815_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5063
p=0.2816
p=0.2617
p=0.0741
p=0.1484
p=0.0343
p=0.1381
p=0.9706
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

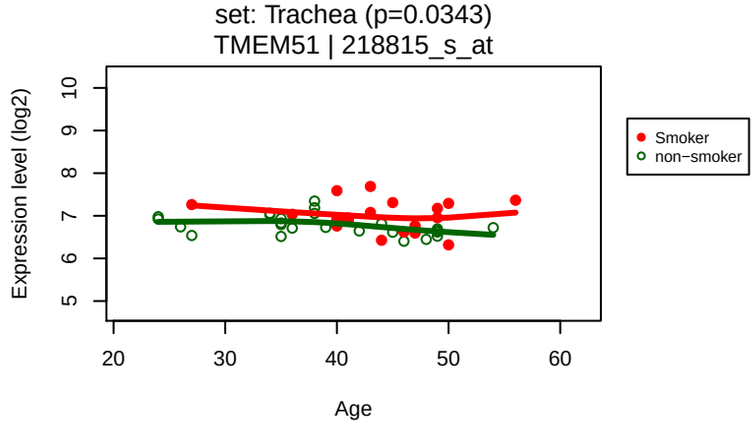

Gene expression ratio in Males
gene: TMEM51 | 218815_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1628
p=0.0402
p=0.2673
p=0.422
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM51 | 218815_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8379
p=0.338
p=0.0386
p=0.4986
SmallAirway2
LargeAirway2
Trachea
Macrophage2

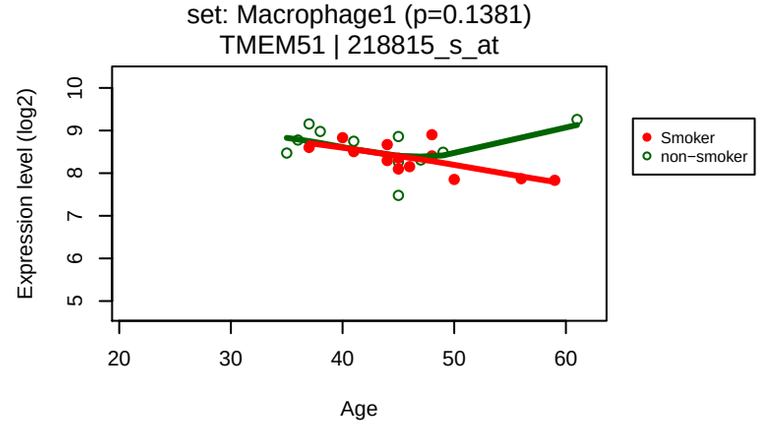

set: SmallAirway1 (p=0.5063)
TMEM51 | 218815_s_at
Expression level (log2)
Age
Smoker
non−smoker

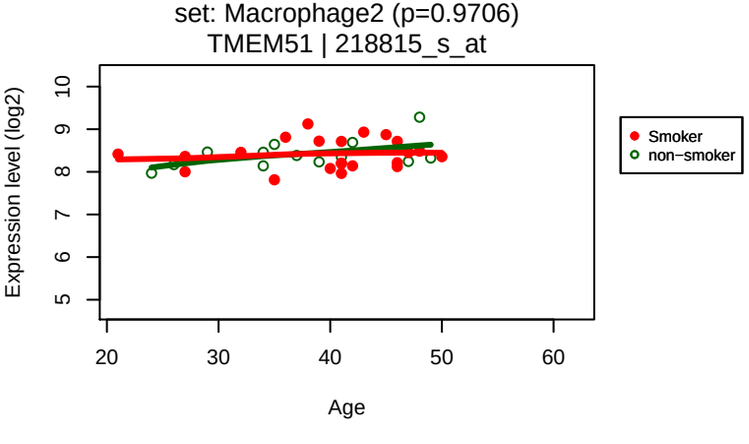

set: SmallAirway2 (p=0.2816)
TMEM51 | 218815_s_at
Expression level (log2)
Age
Smoker
non−smoker

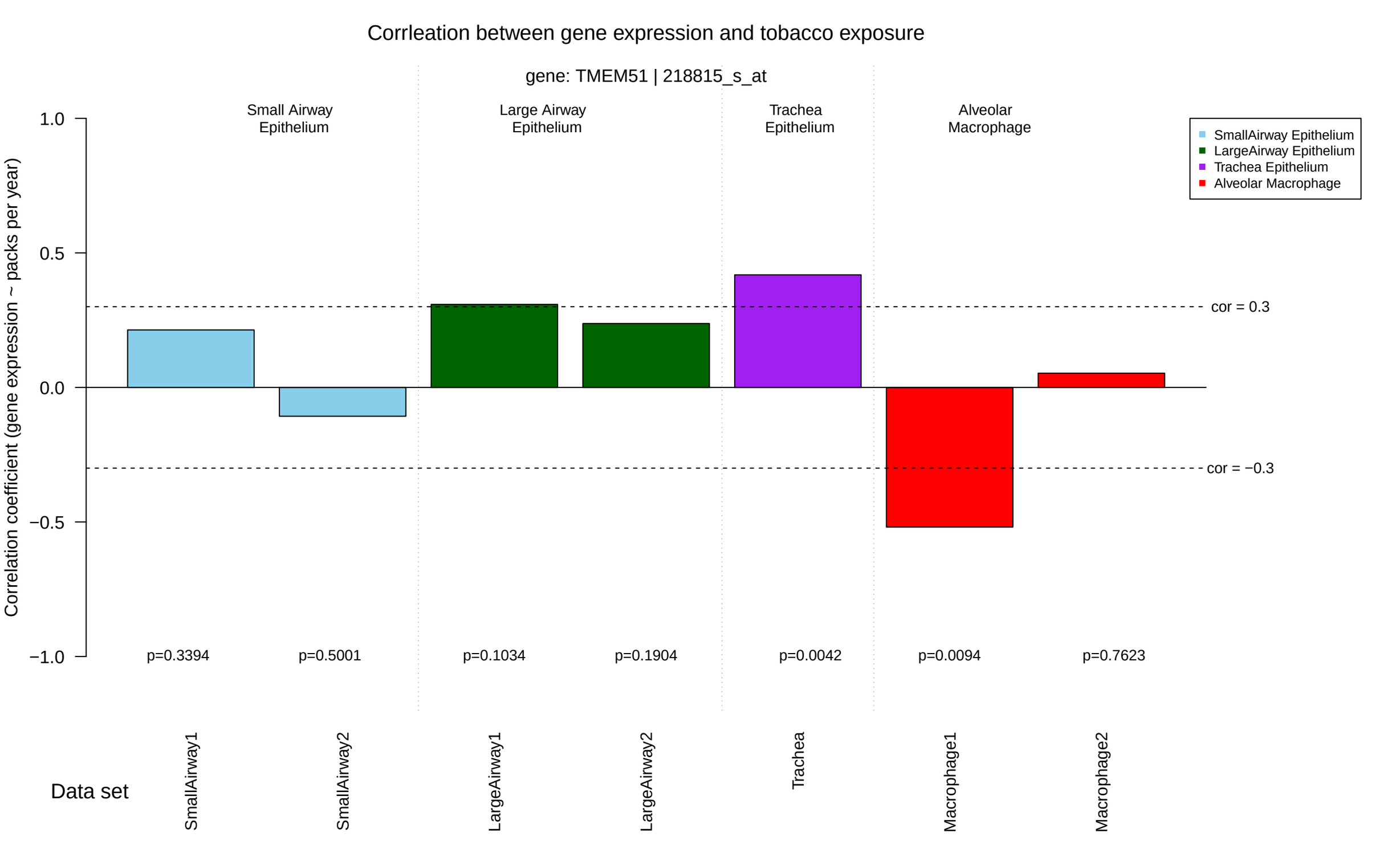

set: LargeAirway1 (p=0.0741)
TMEM51 | 218815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1484)
TMEM51 | 218815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0343)
TMEM51 | 218815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1381)
TMEM51 | 218815_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9706)
TMEM51 | 218815_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM51 | 218815_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3394
p=0.5001
p=0.1034
p=0.1904
p=0.0042
p=0.0094
p=0.7623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2